

THA TUM, Thailand

WMO: 484160

Lat: 15.316N

Lon: 103.680E

Elev: 424

StdP: 14.47

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.4	61.8	50.7	55.7	64.7	53.2	61.1	66.6	13.1	74.2	10.8	72.9	1.5	50	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.8	100.4	78.6	98.5	78.6	96.6	78.6	82.3	90.6	81.7	90.2	81.2	89.8	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	160.6	86.1	79.8	156.9	85.8	79.1	153.6	85.4	46.5	90.5	45.9	90.3	45.4	90.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.1	7.6	55.5	102.8	2.2	2.2	53.9	104.4	52.6	105.6	51.3	106.9	49.7	108.4		
			51.6	83.9	2.7	1.1	49.7	84.8	48.1	85.4	46.6	86.0	44.7	86.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	76.1	80.1	84.7	87.3	86.3	85.7	84.4	83.8	83.0	82.0	79.6	76.0
	DBStd	5.04	4.51	5.18	4.66	3.71	2.81	2.24	2.29	1.95	2.03	2.05	3.70	4.63
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD50	11834	808	842	1075	1119	1127	1072	1066	1049	989	993	889	805
	CDD65	6360	343	422	610	669	662	622	601	584	539	528	439	341
	CDH74	72630	3056	4571	7609	8887	8427	7759	7102	6743	5891	5548	4135	2902
	CDH80	32593	1199	2194	4067	4932	4243	3639	2991	2672	2102	1974	1552	1028
Wind	WSAvg	1.8	1.9	1.5	1.5	1.5	1.6	1.9	1.9	1.7	1.2	1.6	2.5	2.5
Precipitation	PrecAvg	54.5	0.4	0.6	1.9	3.3	6.8	7.7	8.8	9.3	10.4	4.7	0.9	0.1
	PrecMax	69.8	3.2	3.7	7.2	8.1	11.7	12.4	15.0	15.1	20.2	10.8	4.4	0.7
	PrecMin	44.4	0.0	0.0	0.0	1.4	3.4	3.2	1.8	2.4	4.7	0.0	0.0	0.0
	PrecStd	6.4	0.8	0.8	1.7	1.8	2.4	2.6	3.4	3.1	3.5	2.7	1.1	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.9	98.9	102.0	103.7	100.9	98.0	96.6	94.3	93.1	92.2	94.4	92.7
		MCWB	74.7	77.0	77.2	78.5	80.2	80.0	79.7	80.4	80.1	79.0	78.3	76.5
	2%	DB	91.4	96.2	99.6	101.1	98.2	95.8	93.8	92.3	91.3	90.4	91.8	90.2
		MCWB	74.5	76.7	77.6	79.0	80.0	80.0	80.1	80.0	80.0	78.6	77.7	75.6
	5%	DB	89.2	93.9	97.6	98.9	96.0	93.9	92.1	90.9	90.0	89.2	89.9	88.2
		MCWB	74.4	76.2	77.5	78.8	79.8	80.0	80.0	79.8	79.8	78.3	77.2	74.7
	10%	DB	86.8	91.3	95.2	96.6	93.9	92.3	90.4	89.6	88.5	88.0	87.9	85.9
		MCWB	73.5	75.7	77.5	78.7	79.8	79.9	79.8	79.8	79.5	78.1	76.2	73.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.3	81.2	81.7	82.5	82.6	82.6	82.8	82.8	82.6	81.9	81.1	79.9
		MCDB	87.3	92.2	91.7	94.9	92.9	91.3	90.1	89.3	88.8	87.7	88.6	86.7
	2%	WB	77.7	79.5	80.4	81.2	81.7	81.7	81.7	81.8	81.6	80.9	79.9	78.3
		MCDB	86.8	91.2	92.8	93.8	92.1	90.7	89.5	88.5	88.0	87.1	87.7	86.2
	5%	WB	76.1	78.1	79.4	80.5	81.0	81.0	80.9	81.0	80.9	80.1	78.9	76.6
		MCDB	86.0	89.7	92.4	93.1	91.5	90.2	89.0	88.0	87.4	86.3	86.6	85.4
	10%	WB	74.5	76.9	78.5	79.7	80.5	80.6	80.3	80.5	80.2	79.5	77.8	75.1
		MCDB	84.2	88.0	91.6	92.2	90.8	89.8	88.3	87.5	86.4	85.5	85.1	83.6
Mean Daily Temperature Range	5% DB	MDBR	21.2	21.6	19.8	17.8	15.0	13.4	12.3	11.6	11.1	12.2	15.6	18.7
		MCDBR	22.2	22.8	21.7	20.0	17.6	15.5	14.4	13.4	13.0	13.5	17.3	19.9
		MCWBR	8.9	7.8	6.2	5.1	4.8	4.5	4.8	4.8	4.9	5.4	6.5	8.1
	5% WB	MCDBR	20.0	19.9	19.3	18.1	15.5	14.0	13.1	12.4	12.0	12.2	15.2	18.3
		MCWBR	10.1	8.4	7.2	6.0	5.4	5.1	5.3	5.3	5.2	5.6	7.0	9.5
Clear-Sky Solar Irradiance	taub		0.488	0.608	0.668	0.612	0.508	0.469	0.462	0.464	0.503	0.524	0.475	0.466
	taud		2.057	1.763	1.674	1.839	2.162	2.282	2.302	2.287	2.167	2.079	2.192	2.156
	Ebn at Noon		254	229	220	233	254	262	264	266	257	248	256	257
	Edh at Noon		51	72	80	68	49	43	42	43	49	52	44	45
All-Sky Solar Radiation	RadAvg		1592	1670	1693	1814	1800	1695	1609	1548	1491	1596	1578	1539
	RadStd		84	75	65	74	95	71	109	102	109	107	63	43

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	N/A	N/A	+0.58	+0.68	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	+0.49	N/A	N/A	N/A	+0.41	+0.38	N/A	N/A	+189	+189

Nomenclature: See separate page